

Temperature Dependence and Uniformity of Wet Thermal Oxidation of Silicon

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Process Characterization of Wet Thermal Oxidation of Silicon

Abstract

Silicon wafers were thermally oxidized in wet (steam) ambient at two setpoints, 900 °C and 1100 °C, for 60 minutes at temperature, and the resulting SiO₂ films were measured by ellipsometry at five sites per wafer. Mean oxide thickness increased from 909 Å to 4160 Å (4.6×) between the two conditions, corresponding to an apparent activation energy of 1.06 eV, while within-wafer non-uniformity fell from 11.9% to 1.2%. Results are compared to the Deal–Grove model.

Experimental Method

Two sets of Si wafers were oxidized in a resistively-heated tube furnace under a steam ambient. Wafers were loaded and unloaded at approximately 600 °C and the furnace ramped to the process setpoint; the oxidation time (60 min) is defined as the soak duration at setpoint. Post-oxidation film thickness was measured on a reflectometer/ellipsometer at five sites per wafer (center and four edge points). Each fit was recorded with a goodness-of-fit (GOF) value. Data were reduced with a Python/NumPy script that rejects fits outside $0.98 \leq \text{GOF} \leq 1.0$ and computes per-wafer means, within-wafer non-uniformity (range/2·mean), wafer-to-wafer variation, and the temperature dependence of growth.

Results

Metric	900 °C	1100 °C
Valid measurements (rejected)	59 (21)	59 (21)
Mean thickness (Å)	909	4160
Overall std (% of mean)	17.0	1.8
Within-wafer non-uniformity (%)	11.9	1.2
Wafer-to-wafer variation (%)	14.6	1.4

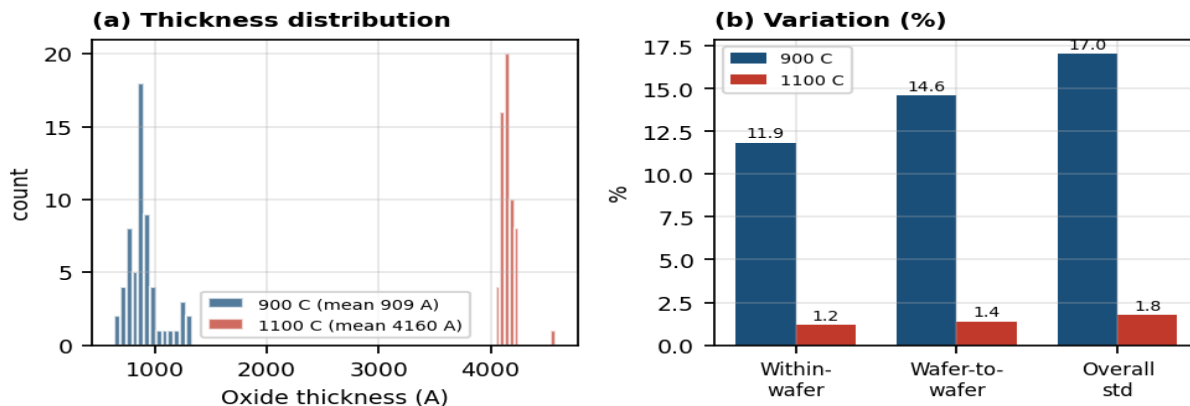


Figure 1. Thickness distributions, uniformity metrics, wafer-mean stability, and the two-point temperature dependence for the 900 and 1100 °C oxidations.

Discussion

Growth and uniformity. The 4.6× thickness increase with a 1.06 eV apparent activation energy confirms the strongly thermally-activated nature of oxidation. The order-of-magnitude improvement in within-wafer non-uniformity (11.9% → 1.2%) is consistent with the transition from reaction-rate-limited growth at 900 °C, to diffusion-limited growth at 1100 °C, where the more uniform diffusion of oxidant through the film governs the rate.

The 900 °C mean agrees with the Deal–Grove steam model to within approximately 5%, whereas the 1100 °C mean falls roughly 30% below the tabulated prediction. Incremental simulation of the load/unload ramps shows that ramp-phase oxidation *adds* film and therefore cannot account for a deficit; the measured parabolic rate constant is approximately 33% of the tabulated value, indicating a furnace water-vapor pressure below the 640 Torr assumed by the standard constants. Because high-temperature growth is diffusion-limited, a reduced oxidant supply suppresses the 1100 °C result most.

Conclusion

Wet oxidation produced 909 and 4160 Å films with a 1.06 eV apparent activation energy and markedly better uniformity at higher temperature; the data match Deal–Grove at 900 °C, and the 1100 °C deviation is attributed to a below-standard water-vapor pressure rather than ramp or measurement effects.